Ideation Phase

Define the Problem Statements

Date	27 October 2026
Team ID	LMP-EV-2026
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Individual project declaration: This project was completed independently by Yuvraj Shrivastava. He independently performed ideation, dataset cleaning and review, Tableau worksheet and dashboard creation, completion of Story Point 1, and website publication. Story Points 2-4 remain planned and were not completed as of 27 October 2026. No other person was involved as a project contributor.

Customer Problem Statement Template:

Create problem statements to understand the customer's point of view. The template helps the individual project creator focus on the customer needs and challenges that the proposed solution is intended to address.

A well-articulated customer problem statement helps identify the challenges customers face and provides a clear basis for designing and evaluating a solution.

I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An EV market strategy director. Emily Johnson is the persona used in the project story, not a confirmed interview participant.	understand how EV sales, category performance, manufacturer contribution, battery technology and manufacturing geography compare so that strategic areas for further attention can be identified.	the information is spread across several measures and views. Comparing category sales, manufacture r sales, battery mix, median range, vehicle counts by country and price/range positioning takes effort.	a decision-maker needs to consider commercial, technical and geographic indicators together. Looking at only one KPI or chart can give an incomplete picture of the market.	less confident about which vehicle categories, manufacturers, regions or battery technologies should receive closer strategic review.
PS-2	An EV product or market analyst comparing vehicle categories and battery configurations.	evaluate product positioning by comparing average range, average price, annual sales, customer rating and battery capacity across vehicle categories and battery types.	the relationships between these measures are not obvious when the analyst views each metric separately, and the data spans multiple vehicle categories and manufacturin g countries.	a product's competitiveness cannot be understood from range, price or customer rating alone. The analyst needs a consolidated and filterable comparison to examine trade-offs.	challenged when explaining which vehicle categories or battery configurations appear competitive and when preparing evidence-based recommendations.

How the Tableau project addresses these needs

The project consolidates key views in an EV market dashboard: Executive Summary KPIs; Category Sales; Manufacturer Share (Treemap); Battery Tech Share; Battery Performance; Country vs Battery Matrix; Price vs Range; Rating vs Battery; and Global Manufacturing Map. The project brief specifies filters for model_year, category, battery_type and manufacturing_country.

Observed evidence from supplied screenshots

The Executive Summary screenshot displays Annual Sales Units of 1,550.70M, Avg. Price of 64.92K, Avg. Range of 325 km and Avg. Customer Rating of 4. The screenshot currently shows a rupee symbol beside the price, but the project owner confirms that the source field price_usd is denominated in US dollars (USD). The dashboard label and display should therefore be corrected to Avg. Price (USD) and $64.92K.

Data preparation and source scope: null values and duplicate records were removed and data types were corrected before analysis. ev_market_train.csv was connected to the Tableau workbook; ev_market_test.csv was supplied as a resource but was not connected. Only Story Point 1, Global EV Market & OEM Overview, is completed; Story Points 2-4 remain planned.

Research boundary: No customer interview transcript or survey evidence was supplied. These statements are project problem hypotheses based on the intended user roles and dashboard capabilities, not verified customer quotations. Emily Johnson is the hypothetical story persona. This project and its documentation were prepared independently by Yuvraj Shrivastava; no other contributor was involved.

Appendix A. Evidence Supporting the Problem Statements

The following screenshots show the dashboard and dataset views relevant to the identified user needs: comparing commercial performance, EV categories, battery technologies, geography, product positioning and customer-rating indicators.

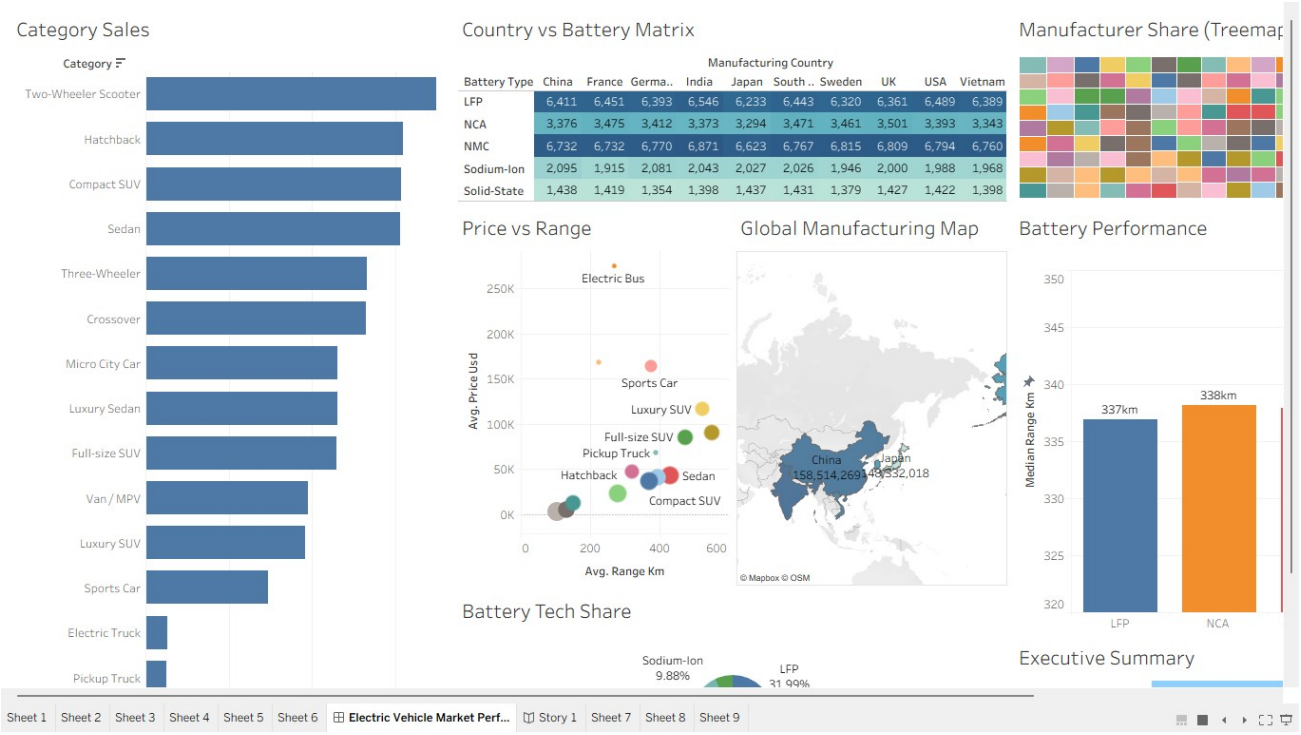


Figure A1. Tableau dashboard overview showing category sales, country-versus-battery matrix, manufacturer treemap, price-versus-range view, global manufacturing map and battery performance.



Figure A2. Executive Summary worksheet showing the displayed annual sales, average price, average range and average customer rating KPIs.

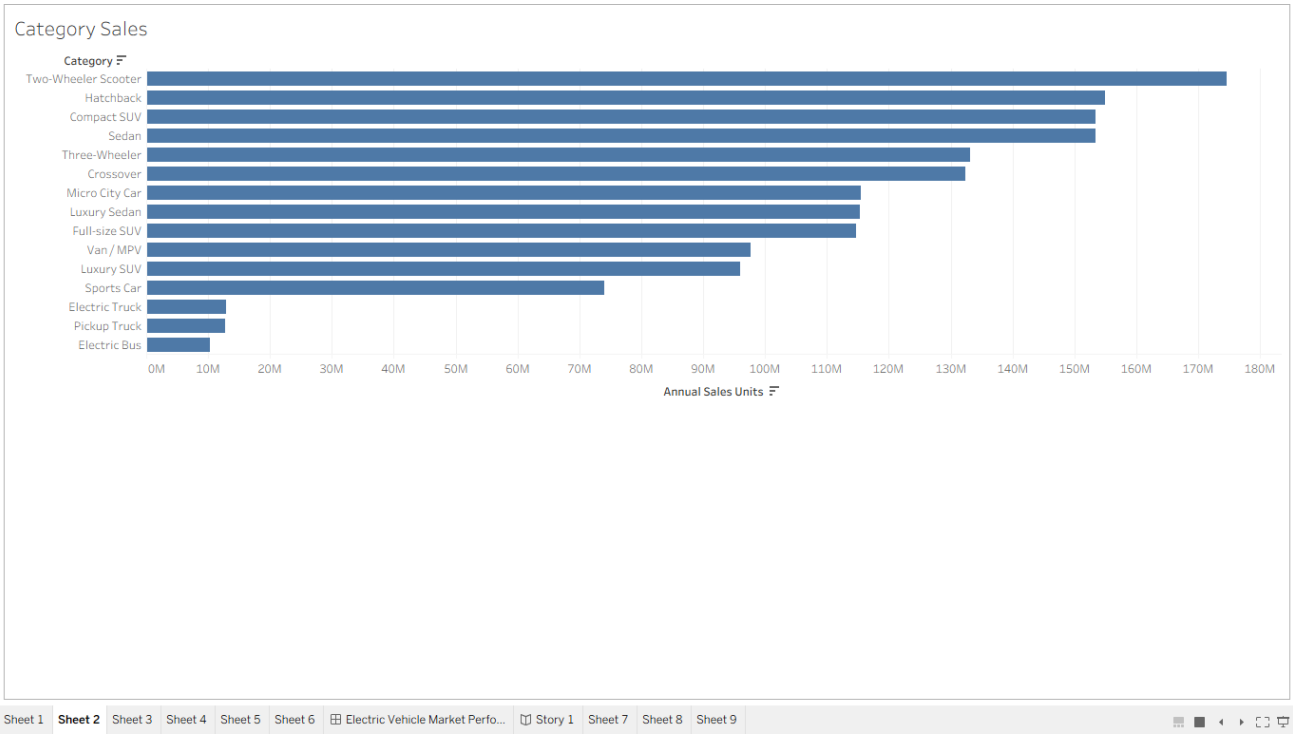


Figure A3. Category Sales worksheet, a horizontal bar chart comparing annual sales units across EV categories.

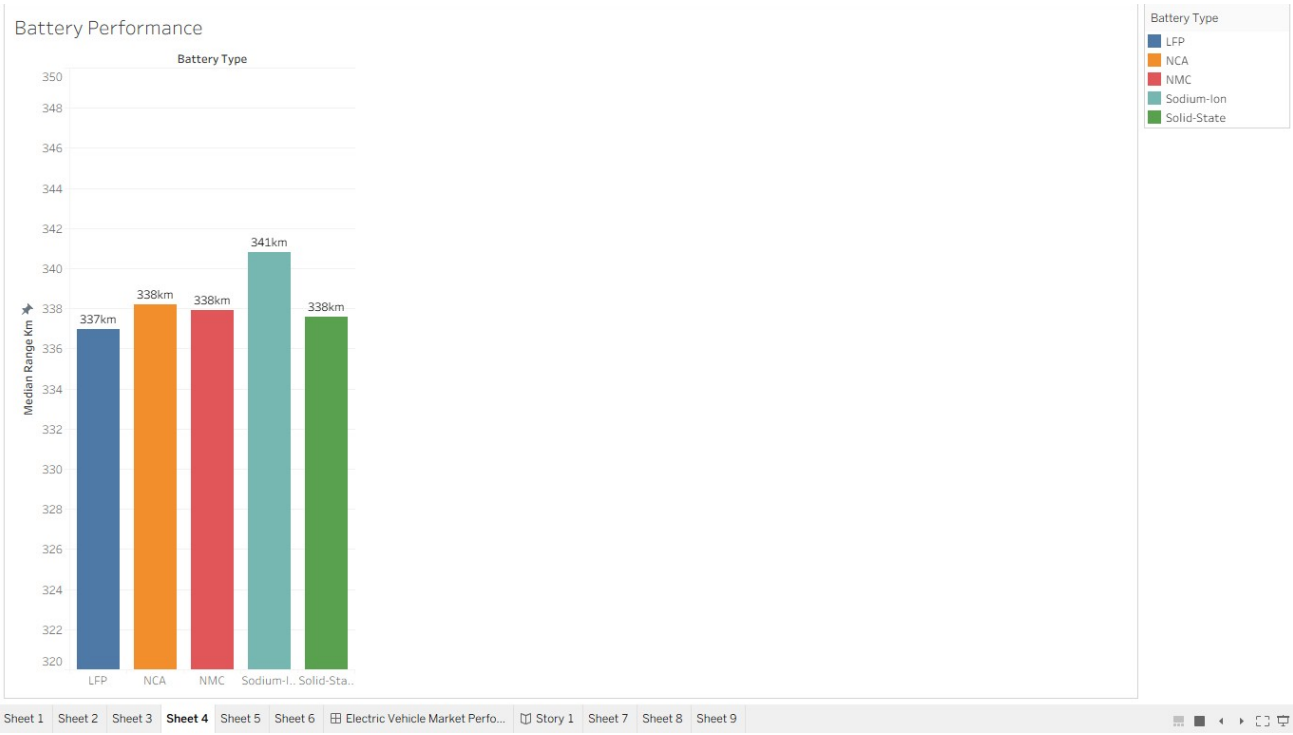


Figure A4. Battery Performance worksheet showing median range values for LFP, NCA, NMC, Sodium-Ion and Solid-State batteries.



Figure A5. Battery Tech Share worksheet showing the displayed percentage mix for battery technologies.

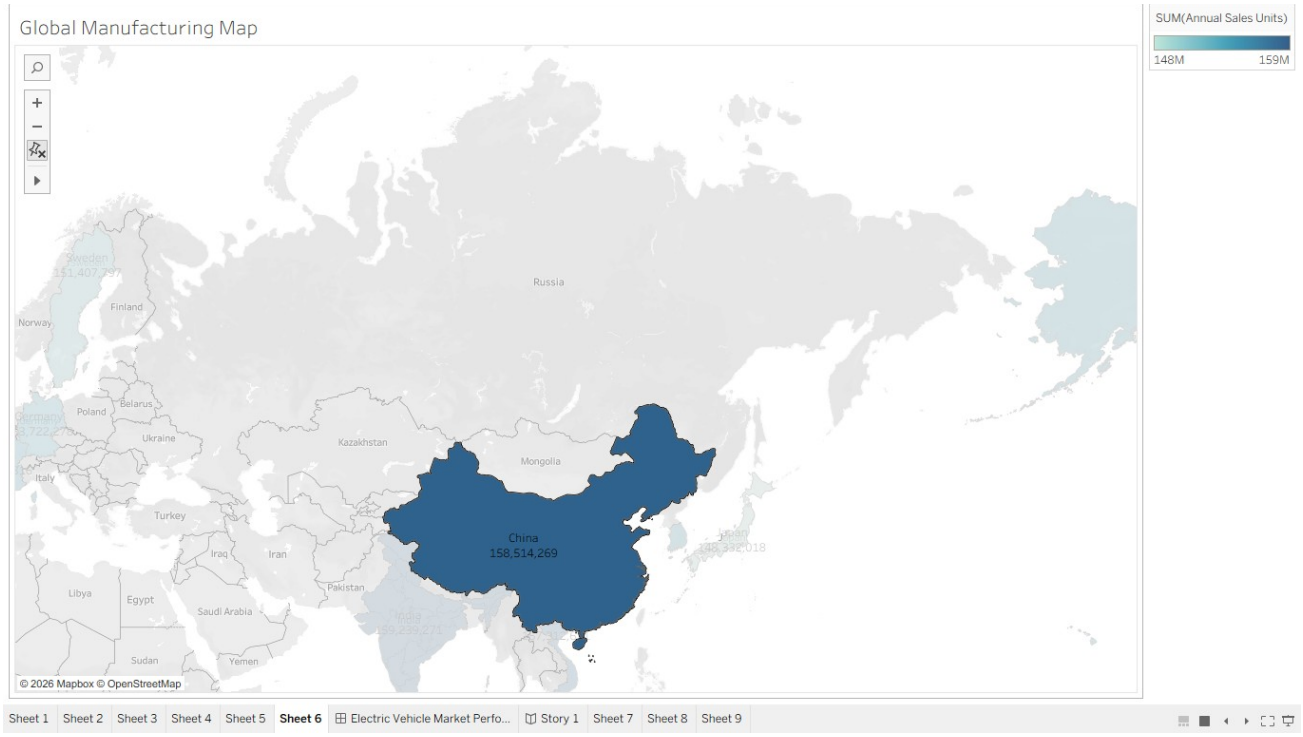


Figure A6. Global Manufacturing Map worksheet, showing geographic sales distribution by manufacturing country.



Figure A7. Price vs Range worksheet, comparing average vehicle range and average price by category.

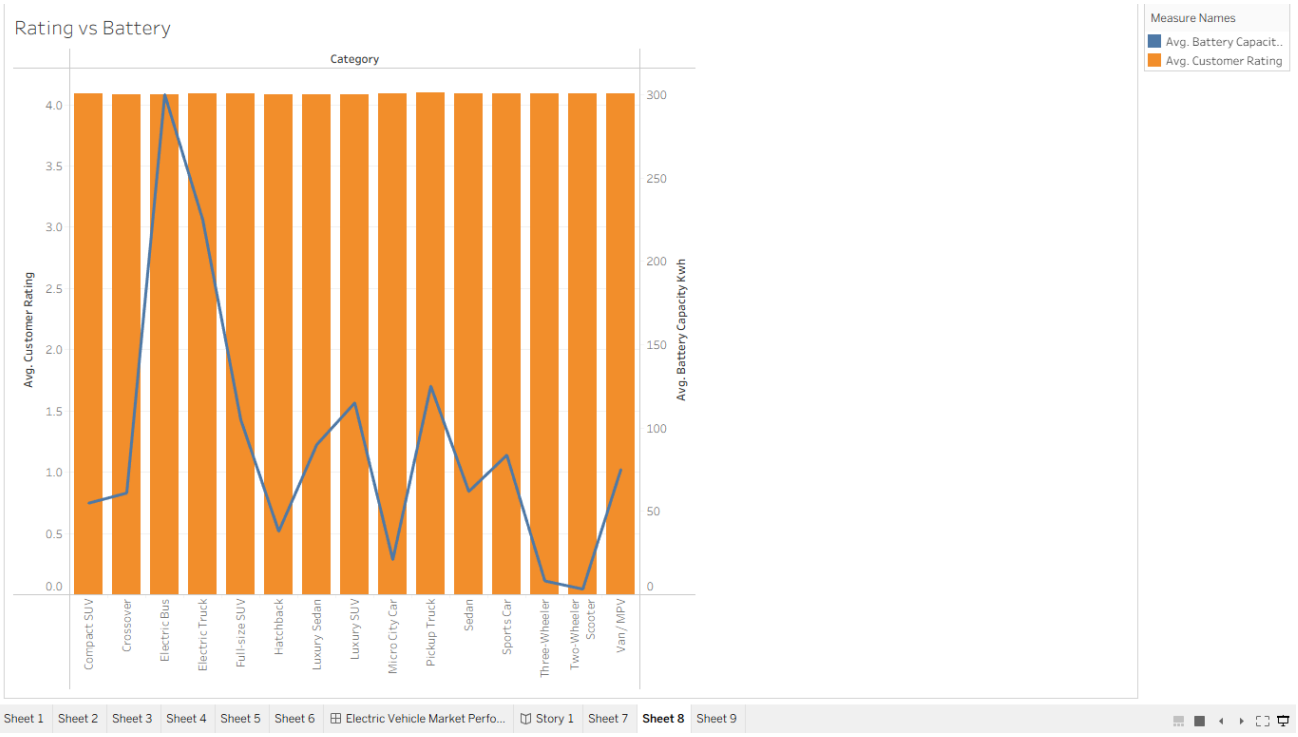


Figure A8. Rating vs Battery worksheet, displaying customer rating bars and a battery-capacity line by vehicle category.

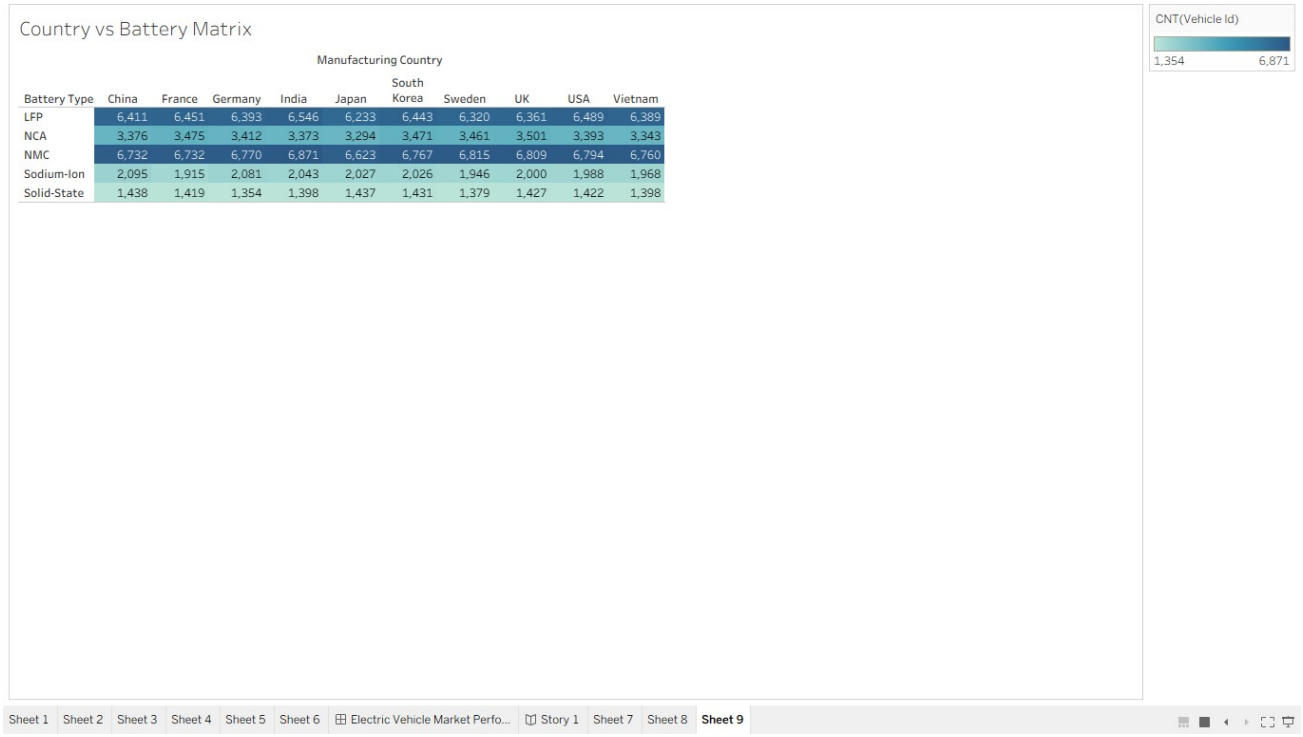


Figure A9. Country vs Battery Matrix worksheet, comparing vehicle counts across battery types and manufacturing countries.

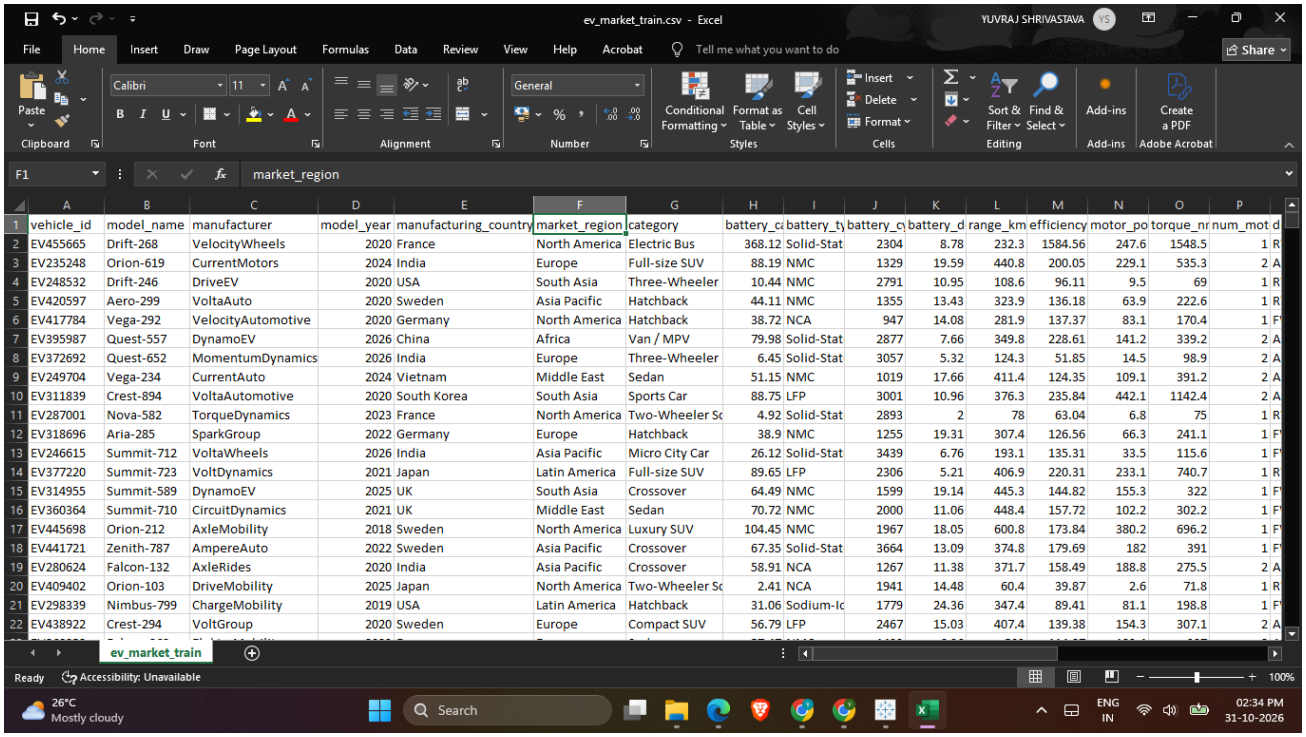


Figure A10. Excel screenshot of ev_market_train.csv, showing sample records and visible dataset fields. It is a sample view, not a complete schema listing.